

COMPUTER SYSTEM (2.0)
QUANTUMTECH
COMPUTING SERVICES
AUTUMN 2024 – WINTER 2025



Name: Min Hein Kyaw

Student ID: 218542

Name of the Centre: Twinkle College



Total word count – 4183 words

Contents

Preface.....	3
Task 1 - QuantumX	4
a) Hardware, computer types and business expansion	4
b) Operating system functionalities and hardware interaction.....	7
Task 2 - Software	10
a) Software types and running pattern in computer	10
b) Development tools	13
Task 3 - Networking	15
a) Network configurations	15
b) LAN topologies	17
c) OSI model.....	19
Task 4 - Future computing services.....	21
a) Distributed computing	21
b) Quantum computing.....	23
Conclusion	24
References.....	25

Preface

QuantumTech is a company which manufactures high-performance computers (QuantumX). It is located in Techville. QuantumX computers have customized operating system with high efficiency. So, demands are high among start-ups and the company also wants to expand his business .

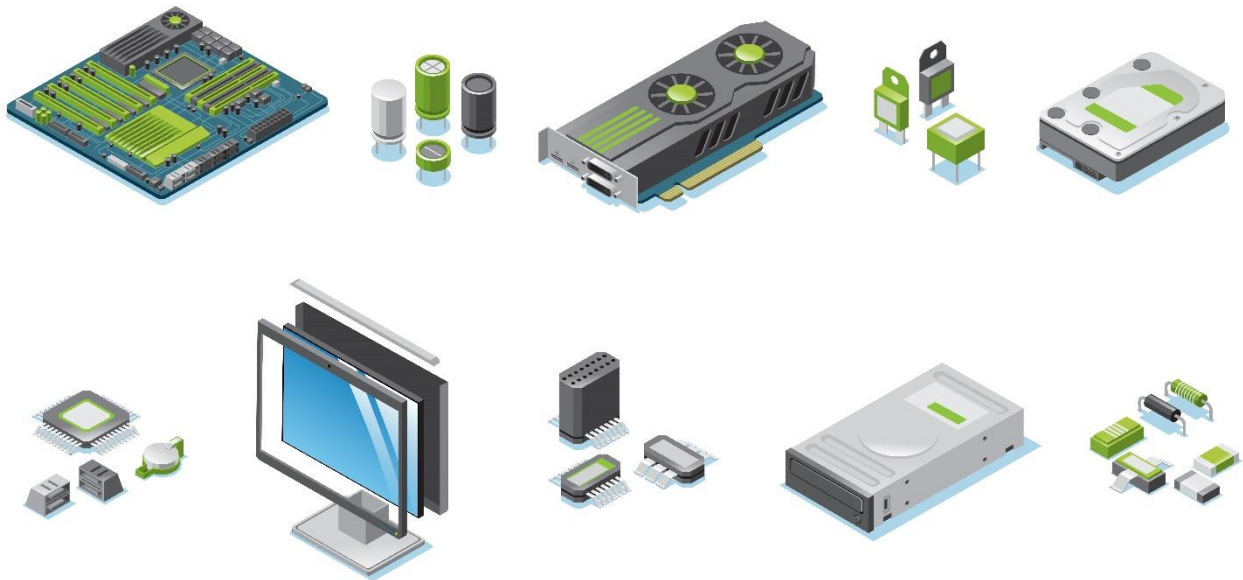


Figure 1. QuantumX (Freepik, 2025)

Task 1 - QuantumX

a) Hardware, computer types and business expansion

Central Processing Unit

It is brain of computer and located on motherboard. It processes instructions by Fetch-Decode-Execute-Store cycle with calculations and decision-making.

Random Access Memory

It is a high-speed, temporary storage for processing data quickly by CPU. It is volatile memory.

Storage (HDD/SSD)

Hard Disk Drive use spinning platters with magnetic coatings. It has mechanical arm in it. It is used to read and write data.

Solid State Drive stores data in flash memory cells electronically. The parts inside are not moving. Although SSD is higher cost and less storage than HDD, it is better performance and more durability.

QuantumX computer

In QuantumX computer, AMD Ryzen 9 9950X3D (CPU) which has Zen 5 architecture, 16/32 cores, large 144MB cache and operation speed (4.3 GHz - 5.7 GHz) is used.

For RAM, G.Skill Trident Z5 Neo which has 64GB capacity (2×32 GB), speed of 5600–6000 MT/s is used. For SSD, Sandisk WD Black SN8100 which has up to 8 TB capacity, reads/write speed (14900/14000 MB/s) is also used.

So, QuantumX is the best performance computer. It can transfer large files with enough storage and also compactable with latest features.

(Loeffler, 2025) (Alcorn, 2025) (Paul Alcorn, 2025)

Mainframe computers

They are very large, powerful and support high storage and processing. They can handle thousands of users and running multiple programs at the same time. They are also high reliability and availability but very expensive. They are used in large organization such as banking and airlines.

Microcomputers

They are small, affordable and suitable for one user. It is easy to use. They are used in general daily purposes such as desktops, laptops, tablets and smartphones. QuantumX computer is one of the types of microcomputers.

Advantages and disadvantages

	Pros	Cons
Mainframe computers	- supports huge tasks and databases - scalability for expansion - high reliability	- expensive - not suitable for individual use - needs special and large environment
Microcomputers	- cheap - user-friendly - portable for carrying - suitable for daily use	- less performance than mainframes - not suitable for large organization.

(geeksforgeeks, 2025)

Businesses expansion

Mainframe systems

Target customers will be large enterprises, data centers and research institutions. Investment will be very high as it needs specialized hardware, software and expertise. It needs to compete with large organization such as IBM and so it has high financial and technical risk.

Network services

Target customers will be microcomputer clients, small and medium sized organizations. It only needs normal equipment and training. It can avoid competing with global mainframe giants. Complementary services can be given together while selling microcomputers.

For QuantumTech company, expansion in network services is suitable as it is already selling microcomputers (QuantumX). It can follow natural extension such as LAN, Wi-Fi, IT infrastructure support. So, it can get recurring services while one-time setups. The worldwide market for microcomputers is growing from \$32.38B(2024) to \$47.17B(2033) at 6.4% CAGR(Compound Annual Growth Rate) and so it is high local demand.

(Adaptiveis.net, 2025)

b) Operating system functionalities and hardware interaction

Device management

OS manages all hardware devices through device drivers. They include printers, GPUs and network cards. It can support QuantumX for smooth operation in between different types of hardware. It reduces system errors and minimizes downtime.

User Interface (GUI/CLI)

OS provides interfaces for users to interact with system. It shows either by GUI(visually) or by CLI(through command). So, technical and non-technical clients can use it for operating system efficiently. It improves productivity and flexibility among different roles.

Networking

OS manages network connections, user access and communication protocols between devices and systems. It can give secure, reliable data communication and can also supports scalability. OS-managed firewalls and encryption protect the users' data and credential information.

(Goyal, 2025)

Hardware interaction

Resource Allocation

OS manages and distributes hardware resources on running applications. They includes CPU time, memory and storage. This supports balanced running time. It prevents conflicts and system overloads. So, users can get maximum performance of QuantumX computer.

Error Handling

OS monitors continuously hardware errors such as hardware failures, memory corruption and device malfunctions. So, if an error is happened, it displays messages on monitor. It can correct automatically, or safely shut down the affected hardware. So, it can protect data and system integrity.

I/O Management

OS controls flow of data between computer and input/output devices. They include printers, keyboard, mouse, and network interfaces. It manages multiple running processes by buffering, queuing requests. So, users and programs can communicate with hardware efficiently.

Device Drivers

OS needs device drivers to communicate with hardware devices because OS cannot directly communicate with them. So, users need to install correct driver for specific hardware to get compatibility, continuous operation and optimal device performance.

(Stephanie Susnjara, 2025)

Alternative characteristics of OS

Multitasking

OS provides running multiple processes at the same time by switching and sharing CPU resources. This improves system performance and productivity of QuantumX.

Security

OS provides user authentication. It also controls user access and provides encryption. It can prevent unauthorized access and malicious attacks and so protect users' data and system integrity.

Memory management

OS allocates and deallocates memory for running processes. It manages both RAM and SSD/HDD by using paging and segmentation techniques. So, it supports optimal performance of QuantumX under heavy workloads.

(geeksforgeeks, 2025)

Total word count Task 1 – 824 words

Task 2 - Software

a) Software types and running pattern in computer

Application software

Users can use directly it to run specific tasks. They include writing, photo editing and browsing websites. It is generally large and user-friendly. High-level languages are used to develop this software. For example - Microsoft Word.

Microsoft Word is an office tool which is used as word processor. Users can write, create, format and edit text documents. It has tools for checking spelling and grammar. It has many templates. Users can create documents such as letters and reports. Then, they can be saved locally or into cloud for sharing with others.

System software

It runs and manages computer hardware. It gives necessary platform. Application software can run onto it. So, It perform as a bridge between hardware software and application software. It is written in low-level languages. For example – Window Operating system (Microsoft Window).

Microsoft Window manages hardware resources. They include storage(SSD/HDD), CPU and RAM. It provides user interface (both GUI and CLI) for users to interact with machine. It handles system recourses and file management. It provides underlying infrastructure for application software. Memory management, input/output management and process management are also handled.

Development software

Programmers use it to create, fix and test error. It maintains other software (application and system software). It has essential tools for software development process. It is not for standard users. For example – Visual studio.

Visual studio has tools for code editing, fixing error, compiling and GUI design. Multiple programming languages such as Python and C++ are supported. It has also version control system. It has compilers that convert written code into executable programs. It can fix error and help code projects management.

(Itesh Sharma, 2021)

Application software

Productivity Software

It is used to help users in managing documents, data and projects by automating repetitive tasks. It is used to manage information such as data within spreadsheets or visual information in presentations. Clients can use it to create, edit, manage various document types. It also supports teams to work together by file sharing. For example – Microsoft Excel.

Multimedia software

Clients can use it to create, edit, manage video and audio files by trimming, cutting, merging and layering (combination). Content can be edited by effects, transitions, filters and noise reduction. It is also used to import, export and convert into various formats. For example – Adobe Premiere Pro.

Communication Software

It is used to connect, collaborate and share information among clients by various formats. It supports real time communication. It is used to send and receive messages. It is also used to call voice and video. It has features for scheduling and reminders. So, it supports remote collaboration in organizations. For example – Zoom.

Entertainment software

It is designed to provide leisure, fun and content consumption. It supports various media and game formats. It supports streaming of movies, music and playback files. It provides real-time gaming experiences by graphics, sound and storylines. It also supports social and multiplayer features such as chat functions. For example – Netflix.

(VP, 2025)

Running pattern of software

Software is written in languages such as Python or C by programmers. These languages need to be converted by compiler or interpreter into machine code (0s and 1s). So, CPU can directly understand and executes (run).

When software is clicked, its program must be loaded from long-term storage (SSD/HDD) into fast temporary memory (RAM). Then, CPU can work with data in RAM.

CPU works with Fetch-Decode-Execute cycle and the cycle is repeated for billions of times per second. CPU uses internal memory units (registers). They are PC(program counter) that holds address of next instruction and IR(instruction register) to manage cycle. CPU fetches next instruction from RAM into IR. It decodes instruction to find out required action such as adding numbers or moving data. Then, it executes action by using AC (accumulator) for calculations.

Data are transferred between CPU and memory by internal connections (buses) such as address bus for specifying location, data bus for carrying content and control bus for indicating operation type (read/write).

For interfacing with other system components, OS (operating system) is important as it acts as a middle man to manage all system resources. It allocates RAM and prevents programs from interfering with each other.

CPU cannot communicate directly with peripheral hardware devices such as keyboard and display. So, when running program wants to use a device, it needs to send a request to OS. Then, OS uses drivers (specialized programs) to handle communication. Drivers translate a request into specific, low-level commands that hardware can understand. So, if running software needs data that is saved on storage (SSD/HDD), OS is responsible for loading that data into RAM where CPU can work with.

(geeksforgeeks, 2025)

b) Development tools

Compiler

It is a specialized program that converts entire source code which is written in high-level (human readable) language such as Java, C into machine code (0s and 1s) which computer can directly execute. This translation occurs before program is executed. It produces files such as .exe or .bin. This process involves many complex steps. They include lexical analysis, optimization, code generation and parsing.

It is essential for creating efficient applications. The machine code runs directly by CPU with optimal performance. If compile-time errors occur, developers can know before program runs. So, developers can find mistakes and improves code reliability and stability.

Interpreter

It executes source code different from compiler as it reads and executes code one statement at one time. So, it translates instructions to run instantly instead of producing an executable file. It is used by languages (for example - Python and JavaScript).

It is good for experimentation, rapid development and debugging. Programmers can test code changes instantly because it gives immediate feedback. So, it is very efficient than using compiler. It is easier to debug programs. If there is an error, programmers can find quickly which line that an error happens. So, it is very useful for scripting, automation and learning new concepts.

Integrated development environment (IDE)

It is a specialized software. It gives a workspace which contains everything for programmer to develop applications. It contains various tools such as code editor, compiler, interpreter, debugger. So, it can use in finding errors, project management and file browsing.

For example – visual studio code (VS Code)

As all tools are available in one place, it improves productivity and reduces complexity. It has features such as real-time error checking , syntax highlighting and auto-filling code. So, programmer can write, test and debug code effectively and efficiently.

Version control system (VCS)

Clients use it to track and manage changes to source file over time. Multiple users can collaborate on same codebase. They can update, merge and revert code files when it is needed. For example – Git (often paired with GitHub)

It is good for teamwork, collaboration and project stability. It can prevent code loss or overwriting because it stores history of project. So, it is safe for experimentation. If errors happen, developers can revert easily to previous version (stable version). So, it helps in reviewing code, resolving conflicts when merging code and improving team collaboration.

(Silveira, 2024)

Total word count for Task 2 – 1184 words

Task 3 - Networking

a) Network configurations

LAN (Local area network)

LAN is designed for small area connection. It connects devices in small area. They include school, home, small building and office. It is used for sharing resources such as files, printers within private network. Cables (Ethernet) or Wi-fi (wireless) are used to connect devices. Central switch or router connects all of the devices. They use private IP addresses and Mac address to find target devices.

So, LAN has smallest range but very high transmission speed with low latency. It has 100 Mbps to 10 Gbps speeds range. It is easy to install and is also cheap. It only uses local hardware and does not need to pay for ISP for connection.

MAN (Metropolitan area network)

MAN is designed for a city or large campus. It is bigger than local area network but smaller than wide area network. Multiple LANs are connected by using MAN across city. It is appropriate for organizations which need to connect multiple buildings across city such as universities. High-speed fiber optic cables or wireless links are used to connect each other.

So, MAN is medium range about 10 to 50 km. It is faster than WAN but slower than LAN (up to 1Gbps). The cost is also higher than LAN but lesser than WAN. It can be private or public.

WAN (Wide area network)

WAN is designed for broad areas covering countries or continents. It is used to connect around the world. The best well-known WAN is Internet. It connects multiple LANs and WANs by using fiber optic cables, telecommunication lines or satellites. It relies on public infrastructure.

So, it has largest range but speed can be lowest than LAN and MAN due to large distance (1Mbps – 1 Gbps). Cost is the highest because it is complex and needs maintenance. It is not usually owned by single organization.

For QuantumTech, LAN is the best choice. It is very suitable for start-up businesses as LANs are the lowest cost solution and easy to set up and maintain. Clients can connect each other locally and only needs WAN when internet is required.

LANs have high speed with low latency. So, clients can share files, printers with high data transfer speed within single building. Management is simple. Security is easy to control with local traffic. LAN can be the strong foundation for future expansion into MANs when business grows. Then, QuantumTech can provide LAN installations services together with sailing computers.

(geeksforgeeks, 2025)

b) LAN topologies

Star topology

In star topology, all devices are connected to central switch or hub. Failure of one device does not impact others and so it has high reliability. But a single point of failure that the whole network shuts down when the switch fails is existed. It is also high scalable. By connecting to central switch, new devices can be added easily. Devices limitation depends on the capacity of central switch.

It is also easy to maintain. Failure devices can be easily detected as all devices are connected to central switch independently. So, isolation of failure devices is also easy and it does not impact on other devices.

In star topology, twisted pair (Ethernet) or fiber optic cables can be used. Each device needs one cable to connect the central switch. Today, switches are used instead of hubs because they are more efficient to get better performance, speed and security. NICs (Network Interface Cards) is necessary for network connection. If it wants to connect internet, routers may be needed. WAPs (wireless access points) are used if star topology is used wirelessly.

Tree topology

It is a hierarchical structure. It combines multiple star topology networks. It has moderate reliability. Failure of device in one branch affects only that branch. But failure of main switch or backbone cable can cause the whole network to shut down. It is highly scalable. It can expand easily by adding branches and sub branches. So, it is suitable for large structured networks.

It is more complex than star topology. Failure of branch is easy to detect and isolate. But the upper tiers have larger impact on whole network.

In tree topology, twisted pair and fiber optic cables can be used depending on size of network. But backbone cable should be fiber optic cable that connects root and sublevels. It requires central main switch or router and secondary switches (for branches). They are arranged hierarchically. NICs are also required for network connection. WAPs can be used in branch for wireless connection.

Mesh topology

In mesh topology, every device connects directly to every other device. So, it is the most reliable topology. Failure of one device does not impact other as there are many alternate paths. It is high redundancy. But scalability is poor because it requires a lot of ports and cables when devices are added. So, it is very expensive for expansion.

Maintenance is complex due to lots of cabling and connections. Troubleshooting can be difficult as there are lots of network traffic in a network.

In full mesh topology, it requires lots of cables as each device connects directly to every other device. Multiple switches or routers are needed to manage traffic. Every device needs NICs to support multiple links. WAPs are used for wireless mesh networks.

Hybrid topology

It includes a mix of two or more topologies such as tree, star and mesh. Reliability is variable as it depends on mixing topologies. It is generally designed for high flexibility by taking the benefit of each topology type. So, it is suitable for large and growing networks.

Complexity depends on mixing topologies and it needs specialist review for troubleshooting and upgrades. So, maintenance can be difficult.

In hybrid topology, a mix of cable types such as twisted pair, coaxial, fiber is used. Fiber optic cable is used for backbone connection. It requires a mix of routers, hubs and switches. Routers are necessary for connecting dissimilar segments. For large hybrids, specialized traffic management hardwares such as load balancers and bandwidth managers are needed. NICs is necessary for both wired and wireless connection. WAPs are needed for segments that use wireless connection.

(Robinette, 2025)

c) OSI model

Quantum X computer has custom OS. But it needs to use TCP/IP model to connect with other devices across a network. TCP/IP is based on OSI 7 layers and it is the global standard of communication across network.

7 layers are existed in OSI model.

Physical layer (Layer1)

It is responsible for transmission of raw, unstructured bits (0s and 1s) over a physical medium such as cables and Wi-Fi (air). It converts data(bits) into electric, light or radio waves. It controls modulation, synchronization and transmission rates.

Data link layer (layer2)

It provides error-free data transfer between devices. It organizes raw bits into frame. It uses MAC addresses to find target device and also handles error detection and correction. Two sublayers such as MAC and LLC are existed.

Network layer (layer3)

It handles logical addressing and routing data between different networks. It uses IP addresses (IPv4/IPv6) to find best routes. Data is carried as packets.

Transport layer(layer4)

It provides complete, reliable and end-to-end data transfer between applications across networks. TCP and UDP protocols are used. TCP is reliable, connection-oriented and accurate. UDP is fast, connectionless and used for real-time applications. They break large data into smaller segments and recombine them at destination.

Session layer (layer5)

It manages sessions between applications. It handles establishment, maintenance and termination of connections. It is also used for authentication and authorization. APIs and NETBIOS are used in this layer.

Presentation layer (layer6)

It is used for translation of data formats and security management. It includes data formatting and code conversion, encryption/decryption and data compression for efficient transfer.

Application layer (layer7)

It is closest to users and delivers services directly by application such as browsers and emails. It acts as window for network access and displays received information to users. HTTP, FTP, SMTP and DNS protocols are used in this layer.

First 3 layers are combined as application layer and last 2 layers are combined as network interface layer in TCP/IP model. By using this model, Quantum X computer is interoperable with different vendors , easy to upgrade and troubleshoot.

(Chrystal R.China, Michael Goodwin, 2025)

Total word count for Task 3 – 1375 words

Task 4 - Future computing services

a) Distributed computing

It is a model that multiple computers (nodes) are connected to act as one unified, powerful computer. It can solve tasks that are very large and complex for single computer by dividing them into smaller subtasks and processing in parallel.

It can increase processing and storage capacity by adding more devices (horizontal scalability). There is no single point of failure as multiple computers are operating tasks together. If one computer fails, others can pick up that computer's tasks to operate. So, downtime is reduced.

Tasks can be split and processed in parallel and so it has faster speed, low latency and better performance. Start-ups can also reduce cost instead of investing in expensive, high-end servers. It also manages data synchronization across all devices to maintain data integrity. But security should be carefully managed because there are many attack-surfaces.

As there are many start-ups who need high performance computing for artificial intelligence, machine learning, cryptography and big data, it is a good project for QuantumTech company.

In distributed computing system, there are two main types of architecture. They are client-server and peer-to-peer.

Distribute computing system architecture

	Client-server	Peer-to-peer
Explanation	- Central server manages requests from multiple clients.	- All nodes are equal - They act as both client and server
Features	- Hierarchical structure - Central server manages all resources	- Decentralized structure - No central control
Advantages	- Easy to manage	- High scalable to get better performance - No single point of failure
Disadvantages	- Single point of failure exists.	- Security management can be harder
Examples	- Email servers	- BitTorrent

QuantumTech company should provide distributed client-server architecture (hybrid) that combines centralized control of client-server with distributed power of peer-to-peer.

Four core characteristics

Concurrency

It means that multiple tasks can be processed simultaneously. In distributed computing , different computers can handle different pieces of tasks simultaneously (parallel processing). So, it can boost performance for heavy workloads such as machine learning and artificial intelligence.

Scalability

More computers can be easily added to expand capacity (horizontal scaling). Then, performance is improved and start-ups can handle more workloads efficiently.

Fault tolerance

If one or multiple computers fail, system can keep running as there are redundancy and backup mechanisms in distributing computing system. Then, start-ups can use without disruption with minimum downtime.

Transparency

Clients can see computer as one single, unified unit although it is distributed with lots of computers and complex in background. Technical details of resource allocation and message-passing are hidden from user and so it is user-friendly (easy to use).

So, it is very useful for new start-ups who are running AI, big data or ML workloads. It can provide necessary huge data and massive resources for AI and ML and parallel processing for big data operation. It can also provide high availability which is important for business continuity.

(Phill Powell, Ian Smalley, 2025)

b) Quantum computing

It is an advanced computer system. Differing from classical computers, it uses qubits (Quantum bits) instead of binary bits that must be 0 or 1. Due to superposition, qubits can exist in state of 0, 1 or both at the same time. Qubits can also influence the state of each other although they are physically far away (Entanglement). So, quantum computer has better computing performance than classical computers.

Quantum computers use interference to amplify correct outcomes and cancel out incorrect outcomes efficiently. External interferences such as thermal noise and vibrations can affect the outcome of qubits (Decoherence). Minimizing them is the great way to get effective quantum computations.

Advantages of Quantum computing

Speed

Complex problems can be solved easily by using quantum computers. They can also search unsorted databases faster than classical computers by using Grover's and Shor's algorithms. They can solve the tasks that take classical computers thousands of years in minutes or hours. So, they are very powerful.

Parallel computing

As qubits can exist multiple states simultaneously, quantum computers can process multiple possibilities at once. Due to superposition and entanglement, they can provide massive parallelism and are much better performance than classical computers.

Optimization

Quantum computers can handle large datasets and complex optimization tasks efficiently. They can be used in supply chain management and financial modeling which need optimal solutions among billions of possibilities. They can find solutions in short amount of time and thus they are very useful for large optimization tasks.

Quantum computers are suitable for cryptography applications. They can either break traditional encryption or create new secure cryptographic systems.

They become threats to current cryptographic standards such as RSA and ECC which based on mathematical difficulty of factoring large numbers. Shor's quantum algorithm can solve these encryptions faster than classical computers and encryption can be broke in short time.

They can provide ultra-secure communication channels by using quantum key distribution (QKD) technique. They can also provide quantum-safe cryptography which can protect both classical and quantum attacks.

Quantum computer can speed-up and discover new patterns in AI training and machine learning. They will be the choice for start-ups who need high performance in these fields. Organizations must prepare for both risks (encryption-breaking) and benefits (quantum-secure systems). Quantum-safe protocols may become essential for finance, healthcare and government sectors in the future.

(David Essex, Alexander S. Gillis, 2025)

Total word count for Task 4 – 800 words

Conclusion

QuantumX computer are built with high-performance resources with customized OS and suitable for start-ups. QuantumTech company have a plan to expand in networking services especially in LAN. It uses TCP/IP which is based on international standard OSI model and so it can communicate with different vendors efficiently.

It has also a plan to make a project of distributed computer system as start-ups need it for high performance computing. In the future, when quantum computing is available, this can become the choice for company to invest in this area.

-----End of the Assignment-----

References

Adaptiveis.net, 2025. *Adaptiveis.net*. [Online]

Available at: <https://adaptiveis.net/blog/network-setup-for-small-business/>
[Accessed 8 10 2025].

Alcorn, P., 2025. *tom's hardware*. [Online]

Available at: <https://www.tomshardware.com/reviews/best-cpus,3986.html>
[Accessed 7 10 2025].

Chrystal R.China, Michael Goodwin, 2025. *IBM*. [Online]

Available at: <https://www.ibm.com/think/topics/osi-model>
[Accessed 28 10 2025].

David Essex, Alexander S. Gillis, 2025. *techtarget*. [Online]

Available at: <https://www.techtarget.com/whatis/definition/quantum-computing>
[Accessed 29 10 2025].

FREEP!K, 2025. *FREEP!K*. [Online]

Available at:

https://www.freepik.com/search?format=search&last_filter=selection&last_value=1&orientation=landscape&query=supply+chain+logistic&selection=1
[Accessed 9 8 2025].

Freepik, 2025. *Freepik*. [Online]

Available at:

https://www.freepik.com/search?format=search&last_filter=page&last_value=4&page=4&query=high+performance+computer&selection=1#uuid=f55e5951-6637-4bb4-be8e-cc1437d009cc
[Accessed 29 10 2025].

geeksforgeeks, 2025. *geeksforgeeks*. [Online]

Available at: <https://www.geeksforgeeks.org/computer-science-fundamentals/classification-of-computers/>
[Accessed 7 10 2025].

geeksforgeeks, 2025. *geeksforgeeks*. [Online]

Available at: <https://www.geeksforgeeks.org/operating-systems/functions-of-operating-system/>
[Accessed 8 10 2025].

geeksforgeeks, 2025. *geeksforgeeks*. [Online]

Available at: <https://www.geeksforgeeks.org/computer-organization-architecture/program->

execution-in-the-cpu/

[Accessed 9 10 2025].

geeksforgeeks, 2025. *geeksforgeeks*. [Online]

Available at: <https://www.geeksforgeeks.org/computer-networks/difference-between-lan-man-and-wan/>

[Accessed 27 10 2025].

Goyal, S., 2025. *unstop*. [Online]

Available at: <https://unstop.com/blog/functions-of-operating-system>

[Accessed 8 10 2025].

Itesh Sharma, 2021. *TatvaSoft*. [Online]

Available at: <https://www.tatvasoft.com/outsourcing/2021/11/what-are-the-3-types-of-computer-software.html>

[Accessed 8 10 2025].

Loeffler, J., 2025. *techradar*. [Online]

Available at: <https://www.techradar.com/computing/best-ddr5-ram>

[Accessed 7 10 2025].

Paul Alcorn, J. W., 2025. *tom's hardware*. [Online]

Available at: <https://www.tomshardware.com/reviews/best-ssds,3891.html>

[Accessed 7 10 2025].

Phill Powell, Ian Smalley, 2025. *IBM*. [Online]

Available at: <https://www.ibm.com/think/topics/distributed-computing>

[Accessed 29 10 2025].

Robinette, D., 2025. *selector*. [Online]

Available at: <https://www.selector.ai/blog/7-network-topologies-pros-cons-and-how-to-design-your-topology/>

[Accessed 27 10 2025].

Silveira, S. F., 2024. *ubiminds*. [Online]

Available at: <https://ubiminds.com/en-us/programming-languages-guide/>

[Accessed 8 10 2025].

Stephanie Susnjara, I. S., 2025. *IBM*. [Online]

Available at: <https://www.ibm.com/think/topics/operating-systems>

[Accessed 8 10 2025].

VP, A., 2025. *toobler*. [Online]

Available at: <https://www.toobler.com/blog/types-of-application-software>

[Accessed 8 10 2025].